

Responsible Software - CheatSheet

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INTRO. Responsible Software. Software engineered in a responsible (<i>ethical</i>) way. 1. With a goal to do good. 2. While preventing avoidable negative impacts. 3. Taking people, other systems, social structures and our planet into account. Active Responsibility. anticipating/preventing negative impacts before occurring. Ethics. Principles/standards that guide human behavior in various contexts. <i>Metaethics.</i> "what does it mean for an action to be right ?" <i>Normative.</i> "how should people act in general (moral principles)?" <i>Applied. (Our approach)</i> "how can i act ethically in a real-world situation?"	
Technical decisions can have <i>ethical</i> implications. Ethical Issue. any situation that may compromise respect of ethical value/principle. Ethical Agency. capacity to take actions that align with ethical values and principles. Ethical Sensitivity. ability to - recognize ethical issues (awareness) - identify their impact: who, what	Ethical Decision Making. 1. Recognize (knowledge+experience+habits) 2. Evaluate (cognitive+reason+ffective) 3. Act. Ethical Lenses. 1. Desirability. product corresponds to what people want. 2. Viability. product generates profits for a business. 3. Feasibility. product can be created with existing or new technology. Fourth criterion (Isaac et al.). Add Ethicality. product should be ethical.
Ethical Dilemmas. decision between multiple alternatives, all conflict with ethical values/principles. <i>None of the options override the other, wrong-wrong choice.</i>	
Moral side. Emotions should be used explicitly/intentionally for responsible design (Roeser). In response to moral issues or motivating immoral behavior. <i>Trolley problem.</i> thought experiment, choice: do nothing → runaway trolley kill people, <u>or</u> intervening → fewer people die. tests which moral principles justify action vs. inaction.	Cognitive side (Normative Ethics). - Utilitarianism(Consequences). choose the greatest good for greatest number - Deontology(Action). do right according to universal principles - Virtue(Person). act as a virtuous person - Care(Relationships). mutual responsibility/ care for each other
Ethical Decision Frameworks. 1. Identify the ethical issues 2. Get the facts 3. Evaluate alternative actions with <i>several ethical theories</i> 4. Choose an option and "test" it 5. Implement your decision and reflect on the outcome.	Example questions for brainstorming - Who will use your product on a daily basis? - Who is in geographical proximity with the users of your product? - Who is in social proximity with the users of your product? - Who is involved in the maintenance of your product? - Who provides services that are needed to run your product? - Who provides funding? - Which communities are active in the application domain?
Stakeholder Individuals/groups/organizations/institutions/societies that might reasonably be affected by technology. Direct Stakeholder. <i>in contact</i> with the product Indirect Stakeholder. affected through <i>consequences of the usage</i> of the software Stakeholder analysis. Identifying every entity that can be affected by software positively/negatively. 1. Brainstorm to find all stakeholders possible using questions. 2. Use Direct vs Indirect Chart to find new stakeholders. Additional Methodologies: <i>Ethix. add unintended stakeholders, Mitchell. add attributes to stakeholders, SAS. use different chart with scale for impact from product and influence on the product.</i>	
Ethical Speculation. How to think, not what to think - anticipate potential impacts on stakeholders. Create a fictional story in two parts. (<i>done at the early design phase</i>) I. Dark and Pessimistic 1. Imagine a character that would be negatively impacted by the software 2. Write a pitch that summarizes their story. 3. Find a title.	 II. Positive outcome 1. Identify 1 or 2 main ethical issues represented in your story. 2. Describe the immediate and future consequences. 3. Imagine a happy ending for character.